

Experiment 4

AIM: Use npm / yarn / pip to install and manage packages

```
C:\Users\chandan>node -v  
v23.5.0
```

```
C:\Users\chandan>npm -v  
10.9.2
```

```
C:\Users\chandan>mkdir npm-lab
```

```
C:\Users\chandan>cd npm-lab
```

```
C:\Users\chandan\npm-lab>npm init -y  
Wrote to C:\Users\chandan\npm-lab\package.json:
```

```
{  
  "name": "npm-lab",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "description": ""  
}
```

```
C:\Users\chandan\npm-lab>npm install
```

```
up to date, audited 1 package in 665ms
```

```
found 0 vulnerabilities
```

```
C:\Users\chandan\npm-lab>npm i axios express
```

```
added 76 packages, and audited 77 packages in 6s
```

```
24 packages are looking for funding  
run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
C:\Users\chandan\npm-lab>npm i nodemon --save-dev
```

```
added 26 packages, and audited 103 packages in 4s
```

```
29 packages are looking for funding  
run `npm fund` for details
```

found 0 vulnerabilities

```
C:\Users\chandan\npm-lab>npm run start
```

```
> npm-lab@1.0.0 start
> node index.js
```

```
Server running at http://localhost:3000
Terminate batch job (Y/N)? y
```

```
C:\Users\chandan\npm-lab>type .gitignore
node_modules/
```

```
C:\Users\chandan\npm-lab>git init
```

```
Initialized empty Git repository in C:/Users/chandan/npm-lab/.git/
```

```
C:\Users\chandan\npm-lab>git add .
```

```
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF
the next time Git touches it
```

```
warning: in the working copy of 'package.json', LF will be replaced by CRLF the
next time Git touches it
```

```
C:\Users\chandan\npm-lab>git commit -m "Add new Content"
```

```
[master (root-commit) 9a1cc2a] Add new Content
```

```
4 files changed, 1360 insertions(+)
```

```
create mode 100644 .gitignore
```

```
create mode 100644 index.js
```

```
create mode 100644 package-lock.json
```

```
create mode 100644 package.json
```

```
C:\Users\chandan\npm-lab>git remote add origin
```

```
https://github.com/chandanktech1/OST-Chandan.git
```

```
C:\Users\chandan\npm-lab>git branch -M main
```

```
C:\Users\chandan\npm-lab>git push -u origin main
```

```
To https://github.com/chandanktech1/OST-Chandan.git
```

```
! [rejected]        main -> main (fetch first)
```

```
error: failed to push some refs to 'https://github.com/chandanktech1/OST-
Chandan.git'
```

```
hint: Updates were rejected because the remote contains work that you do not
```

```
hint: have locally. This is usually caused by another repository pushing to
```

```
hint: the same ref. If you want to integrate the remote changes, use
```

```
hint: 'git pull' before pushing again.
```

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

```
C:\Users\chandan\npm-lab>git pull origin main --rebase
```

```
remote: Enumerating objects: 29, done.
```

```
remote: Counting objects: 100% (29/29), done.
```

```
remote: Compressing objects: 100% (21/21), done.
```

```
remote: Total 29 (delta 7), reused 11 (delta 3), pack-reused 0 (from 0)
```

```
Unpacking objects: 100% (29/29), 2.57 MiB | 1.64 MiB/s, done.
```

```
From https://github.com/chandanktech1/OST-Chandan
```

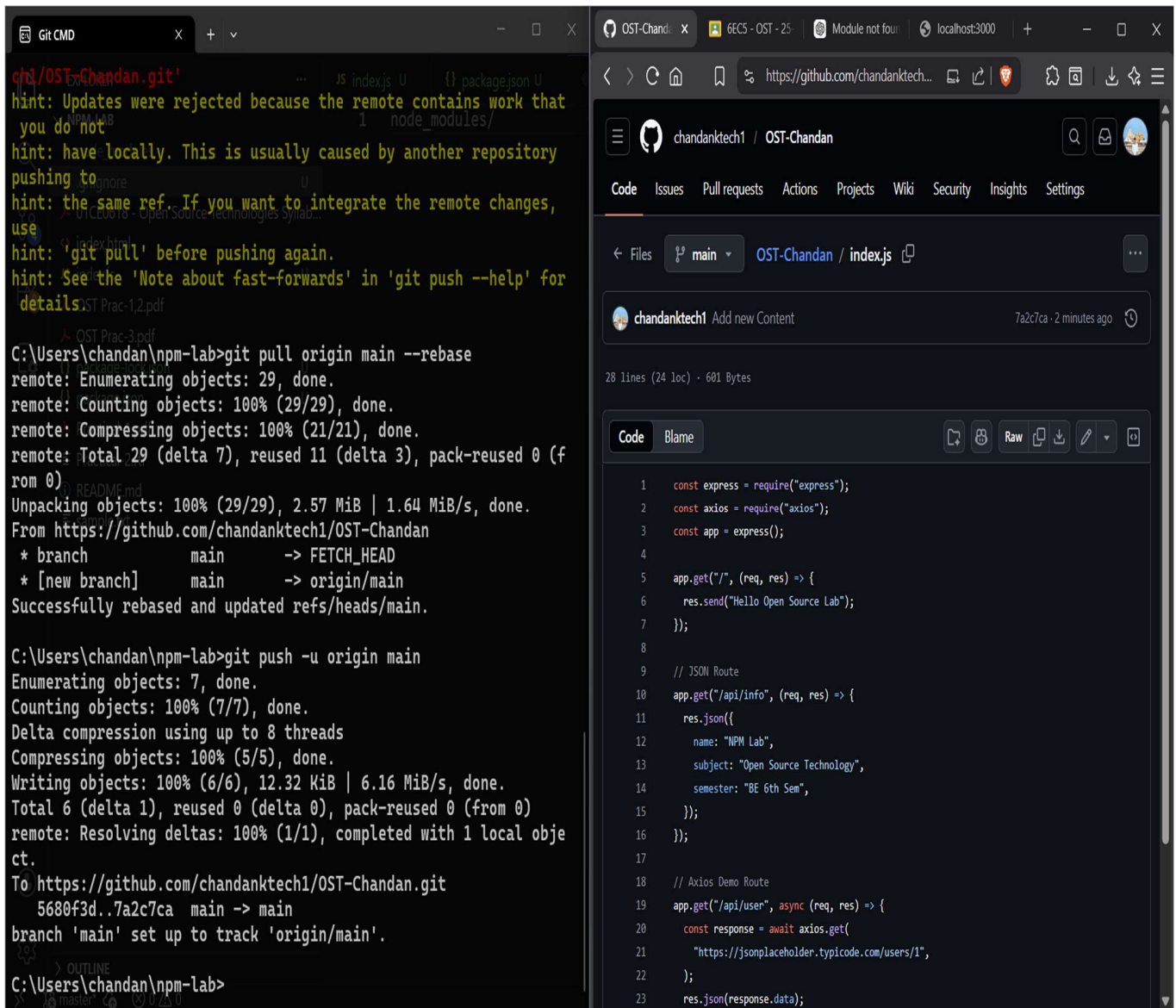
```
* branch          main          -> FETCH_HEAD
```

```
* [new branch]     main          -> origin/main
```

Successfully rebased and updated refs/heads/main.

```
C:\Users\chandan\npm-lab>git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 12.32 KiB | 6.16 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/chandanktech1/OST-Chandan.git
   5680f3d..7a2c7ca  main -> main
branch 'main' set up to track 'origin/main'.
```

```
C:\Users\chandan\npm-lab>
```



The image shows a terminal window on the left and a web browser on the right. The terminal window displays the output of a `git pull` command, which successfully rebasees and updates the local `main` branch. The web browser shows the GitHub repository page for `chandanktech1 / OST-Chandan`, displaying the `index.js` file with 28 lines of code.

```
ch/OST-Chandan.git'
hint: Updates were rejected because the remote contains work that
you do not
hint: have locally. This is usually caused by another repository
pushing to
hint: the same ref. If you want to integrate the remote changes,
use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for
details.
C:\Users\chandan\npm-lab>git pull origin main --rebase
remote: Enumerating objects: 29, done.
remote: Counting objects: 100% (29/29), done.
remote: Compressing objects: 100% (21/21), done.
remote: Total 29 (delta 7), reused 11 (delta 3), pack-reused 0 (f
rom 0)
Unpacking objects: 100% (29/29), 2.57 MiB | 1.64 MiB/s, done.
From https://github.com/chandanktech1/OST-Chandan
* branch      main      -> FETCH_HEAD
* [new branch] main      -> origin/main
Successfully rebased and updated refs/heads/main.

C:\Users\chandan\npm-lab>git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 12.32 KiB | 6.16 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
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ct.
To https://github.com/chandanktech1/OST-Chandan.git
5680f3d..7a2c7ca main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\chandan\npm-lab>
```

The web browser shows the GitHub repository page for `chandanktech1 / OST-Chandan`. The file `index.js` is displayed with the following code:

```
1  const express = require("express");
2  const axios = require("axios");
3  const app = express();
4
5  app.get("/", (req, res) => {
6    res.send("Hello Open Source Lab");
7  });
8
9  // JSON Route
10 app.get("/api/info", (req, res) => {
11   res.json({
12     name: "NPM Lab",
13     subject: "Open Source Technology",
14     semester: "BE 6th Sem",
15   });
16 });
17
18 // Axios Demo Route
19 app.get("/api/user", async (req, res) => {
20   const response = await axios.get(
21     "https://jsonplaceholder.typicode.com/users/1",
22   );
23   res.json(response.data);
24 }
```